

LAB EXPERIMENT RECORD

SET 9 — Online Learning Activity Analysis

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	AKILA S
Register Number	192424331
Date	3 October 2026
Day	Saturday

A. Aim

To analyze student online learning activity and performance using histograms, boxplots, scatter plots, moving averages, and an interactive Tableau dashboard.

B. Dataset

Your pasted dataset formatting is mixed up, but the records appear to be:

Student ID	Gender	Age	Course	Study Time	Videos Watched	Quiz Score	Login Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	SQL	4.2	15	85	2025-01-05
L03	Male	19	R	2.0	8	65	2025-02-08
L04	Female	23	R	5.0	18	92	2025-02-08
L05	Male	21	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

If your actual CSV has slightly different values, use the CSV values in the code rather than manually entering this table.

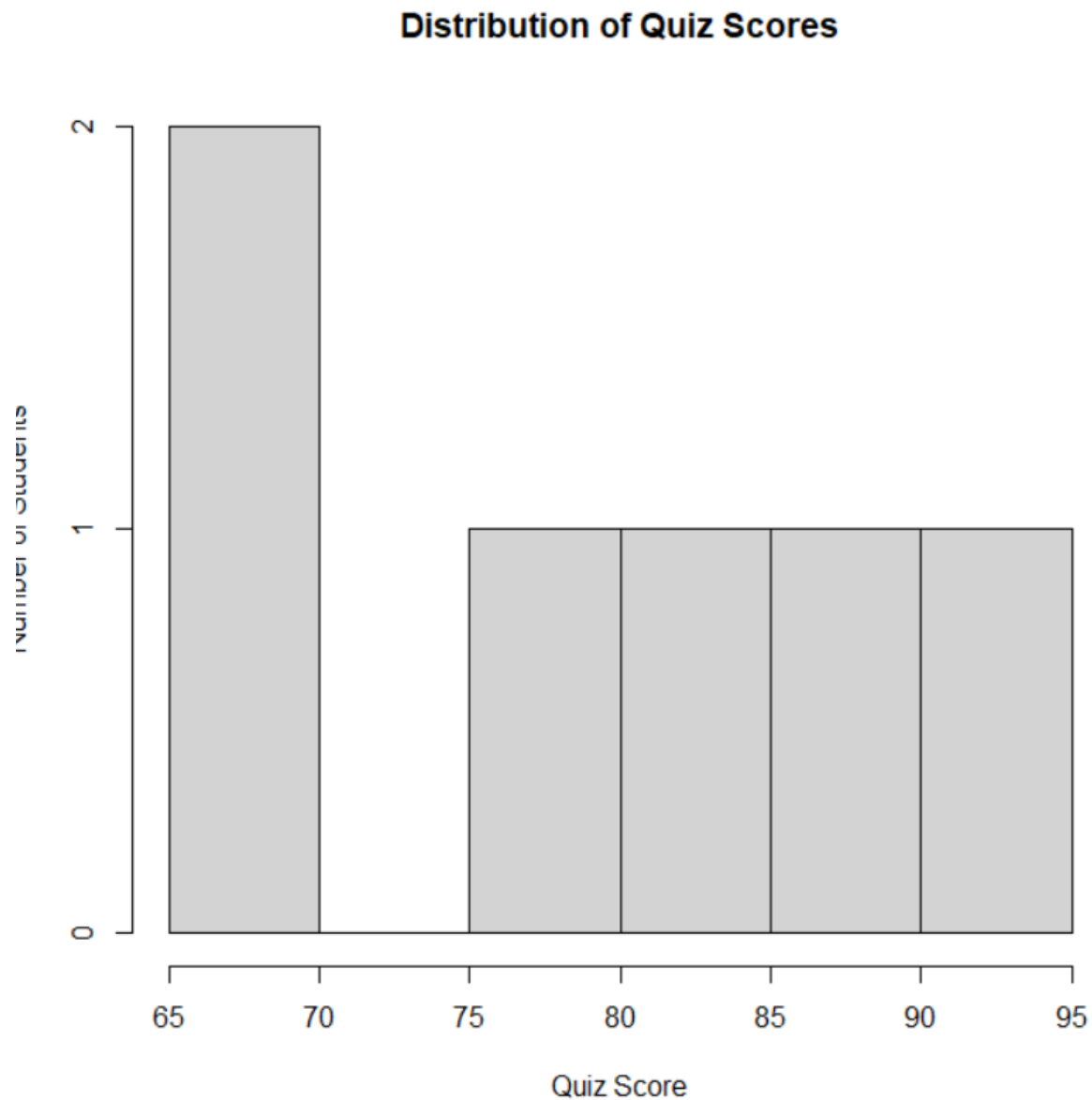
Program 1 — Histogram and Boxplot of Quiz Score

Code

```
data <- read.csv("Online_Learning_Activity.csv")
```

```
hist(  
  data$Quiz_Score,  
  main = "Distribution of Quiz Scores",  
  xlab = "Quiz Score",  
  ylab = "Number of Students"  
)
```

```
boxplot(  
  Quiz_Score ~ Course,  
  data = data,  
  main = "Quiz Score by Course",  
  xlab = "Course",  
  ylab = "Quiz Score"  
)
```



Observation

The histogram shows the distribution of student quiz scores, while the boxplot allows comparison of quiz scores between different courses.

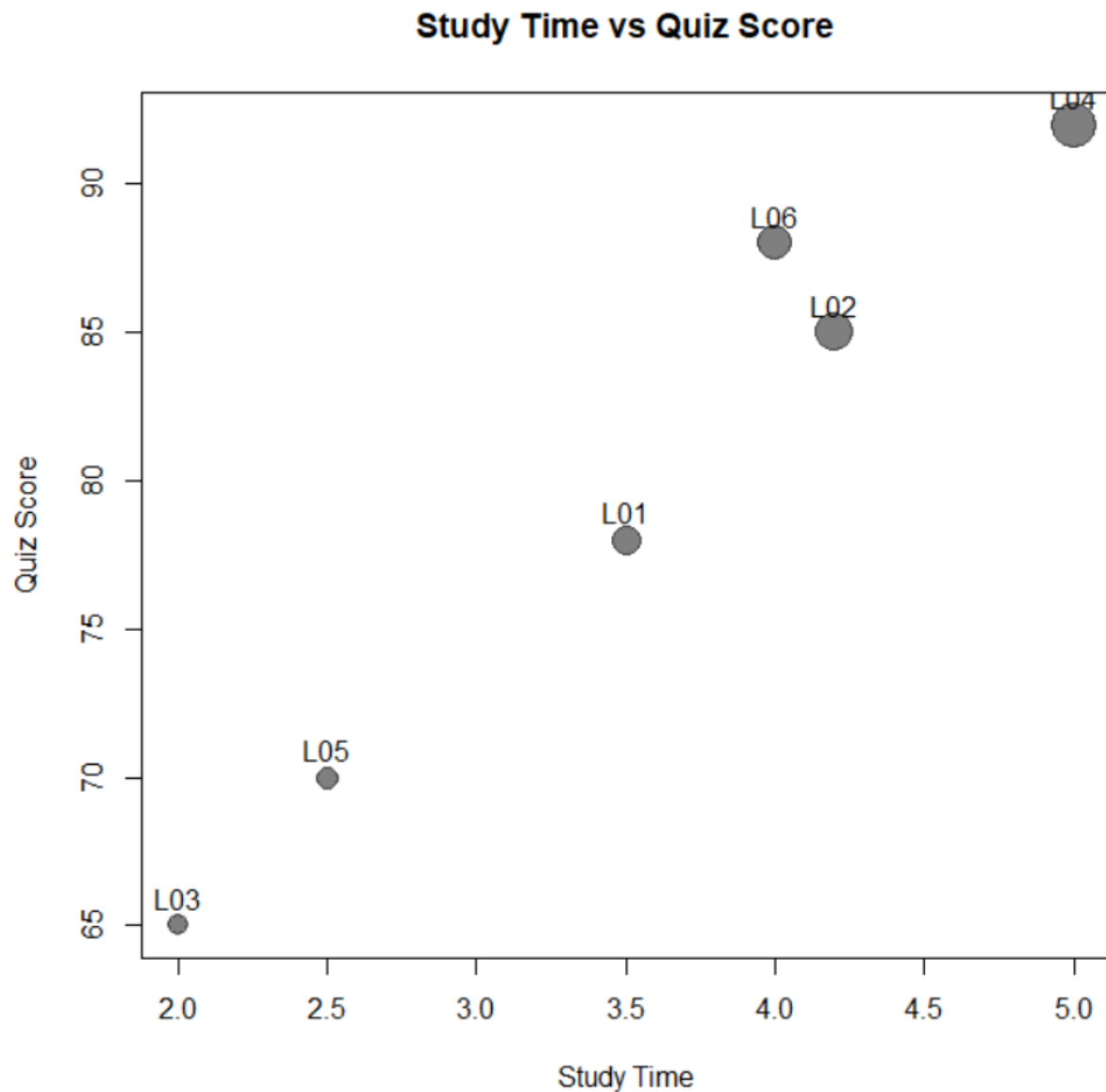
Program 2 — Scatter Plot: Study Time vs Quiz Score

Code

```
data <- read.csv("Online_Learning_Activity.csv")
```

```
plot(  
  data$Study_Time,  
  data$Quiz_Score,  
  cex = data$Videos_Watched / 5,  
  pch = 19,  
  col = rgb(0, 0, 0, 0.5),  
  main = "Study Time vs Quiz Score",  
  xlab = "Study Time",  
  ylab = "Quiz Score"  
)
```

```
text(  
  data$Study_Time,  
  data$Quiz_Score,  
  labels = data$Student_ID,  
  pos = 3  
)
```



Observation

The scatter plot helps identify the relationship between study time and quiz score. Bubble size represents the number of videos watched. Students with greater study time generally show higher quiz scores in the given data.

Program 3 — Monthly Average Quiz Score with Moving Average

Code

```
data <- read.csv("Online_Learning_Activity.csv")
```

```
data$Login_Date <- as.Date(data$Login_Date)
```

```
data$Month <- format(data$Login_Date, "%Y-%m")
```

```
monthly_avg <- aggregate(  
  Quiz_Score ~ Month,  
  data = data,  
  FUN = mean  
)
```

```
print(monthly_avg)
```

```
plot(  
  1:nrow(monthly_avg),  
  monthly_avg$Quiz_Score,  
  type = "o",  
  pch = 19,  
  xaxt = "n",  
  main = "Monthly Average Quiz Score",  
  xlab = "Month",  
  ylab = "Average Quiz Score"  
)
```

```
axis(  
  1,  
  at = 1:nrow(monthly_avg),
```

```

    labels = monthly_avg$Month
  )

  if (nrow(monthly_avg) >= 3) {

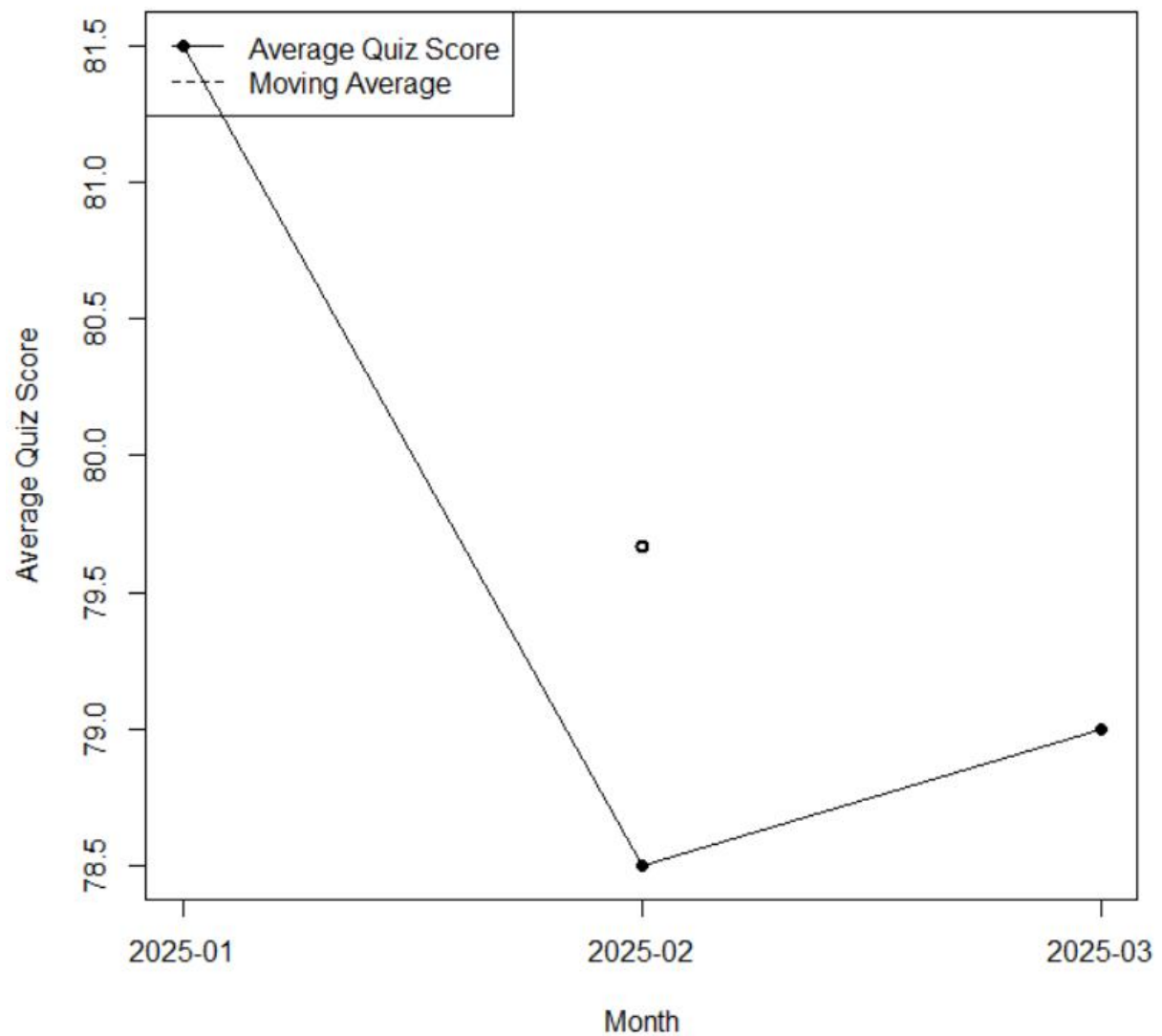
    moving_avg <- stats::filter(
      monthly_avg$Quiz_Score,
      rep(1/3, 3),
      sides = 2
    )

    lines(
      1:nrow(monthly_avg),
      moving_avg,
      type = "o",
      lty = 2,
      lwd = 2
    )

    legend(
      "topleft",
      legend = c("Average Quiz Score", "Moving Average"),
      lty = c(1, 2)
    )
  }
}

```

Monthly Average Quiz Score



Observation

The monthly average quiz score shows the performance trend across different months. The moving average smooths short-term variations and makes the overall trend easier to observe.

Program 4 — Tableau Mini Interactive Dashboard

Step 1 — Import Dataset

1. Open Tableau.
2. Select Connect → Text File.
3. Select Online_Learning_Activity.csv.
4. Check that the columns are correctly recognized.

Step 2 — Average Quiz Score by Course

1. Create a new worksheet.
2. Drag Course → Columns.
3. Drag Quiz_Score → Rows.
4. Change aggregation to Average.
5. Select Bar Chart.
6. Turn on Show Mark Labels.

Step 3 — Study Time vs Quiz Score

1. Create another worksheet.
2. Drag Study_Time → Columns.
3. Drag Quiz_Score → Rows.
4. Drag Videos_Watched → Size.
5. Drag Student_ID → Detail.
6. Select Circle/Scatter Plot.

Step 4 — Add Filters

Add:

- Gender
- Course

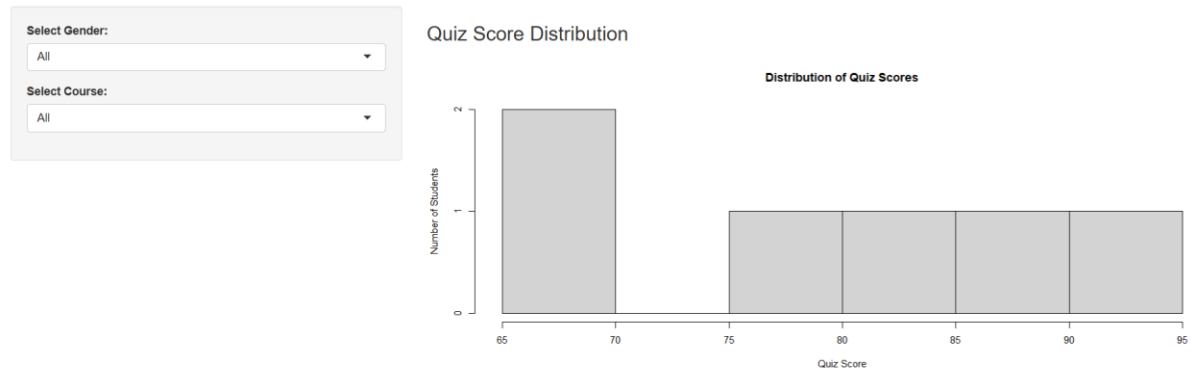
Then select Show Filter.

Step 5 — Create Dashboard

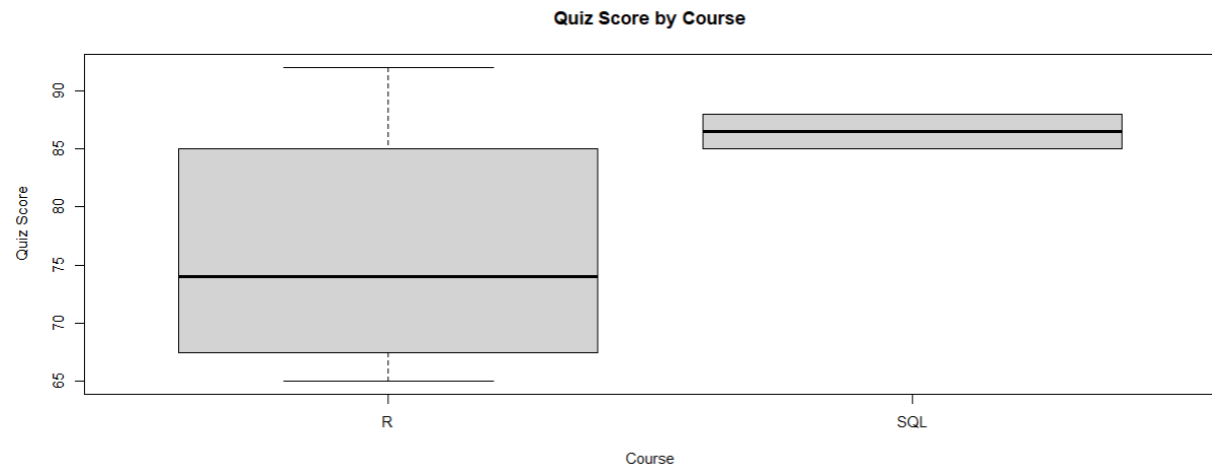
1. Select New Dashboard.
2. Add the Average Quiz Score by Course chart.
3. Add the Study Time vs Quiz Score scatter plot.
4. Add the Gender filter.
5. Add the Course filter.
6. Arrange the charts neatly.
7. Title it:

Online Learning Activity Dashboard

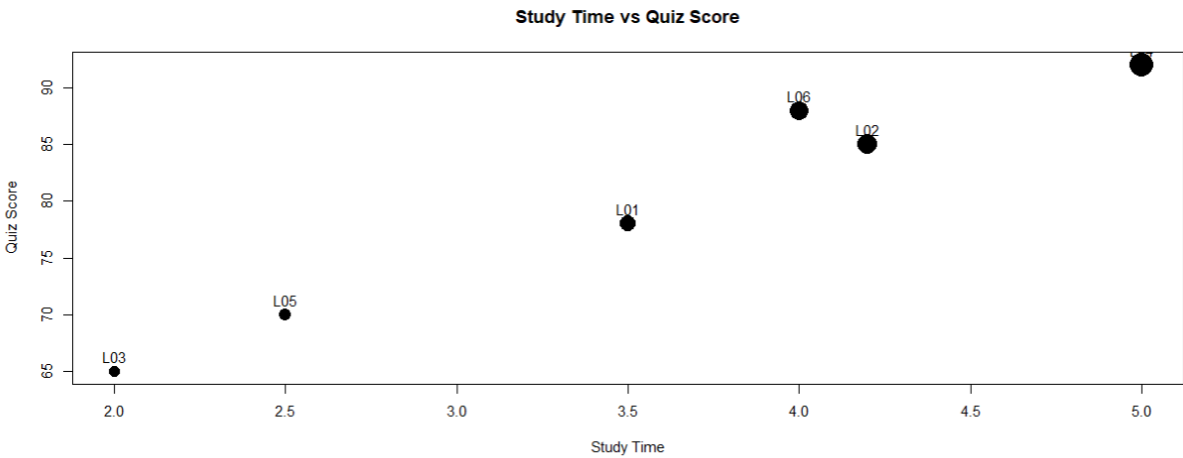
Online Learning Activity Dashboard



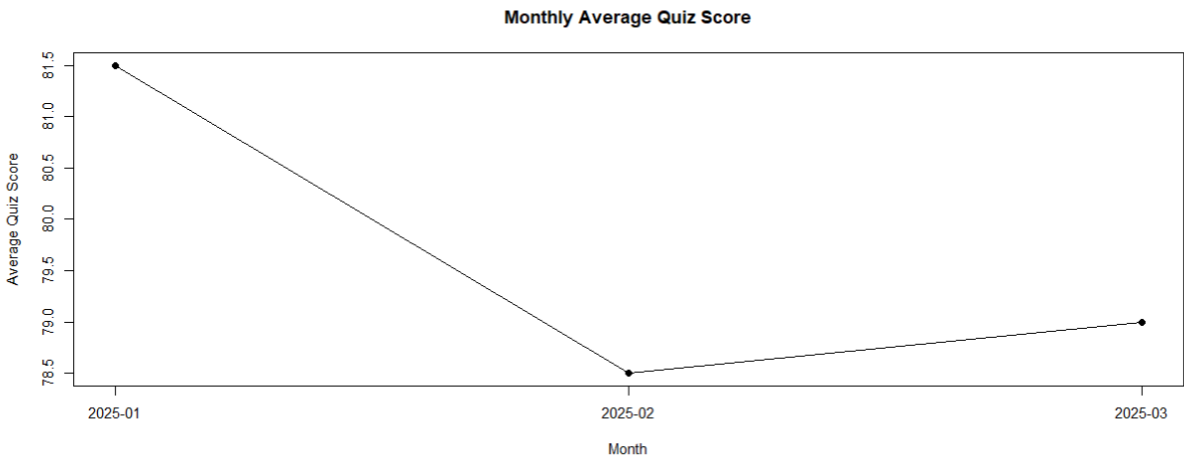
Quiz Score by Course



Study Time vs Quiz Score



Monthly Average Quiz Score



Student Data

Student_ID	Gender	Age	Course	Study_Time	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20.00	R	3.50	12.00	78.00	20093.00
L02	Female	22.00	SQL	4.20	15.00	85.00	20093.00
L03	Male	19.00	R	2.00	8.00	65.00	20127.00
L04	Female	23.00	R	5.00	18.00	92.00	20127.00
L05	Male	21.00	R	2.50	9.00	70.00	20159.00
L06	Female	20.00	SQL	4.00	14.00	88.00	20159.00

C. Overall Procedure

1. Import the online learning dataset into R.
2. Analyze quiz-score distribution using a histogram and boxplot.
3. Analyze study time and quiz score using a bubble scatter plot.
4. Convert login dates into Date format.
5. Calculate monthly average quiz scores.
6. Apply moving-average smoothing.
7. Import the CSV into Tableau.
8. Create course-wise average quiz-score and study-time/quiz-score visualizations.
9. Add Gender and Course filters.
10. Combine the visualizations into an interactive dashboard.

D. Result

The online learning activity of students was successfully analyzed using histograms, boxplots, bubble scatter plots, monthly trend analysis, moving averages, and a Tableau dashboard.

E. Key R Functions Used

Function	Purpose
read.csv()	Import CSV dataset
hist()	Create histogram
boxplot()	Create boxplot
plot()	Create scatter/line plots
rgb()	Apply transparency
as.Date()	Convert date format
format()	Extract month
aggregate()	Calculate monthly average
filter()	Calculate moving average
lines()	Add moving-average line
axis()	Customize axis

legend()	Add legend
text()	Add labels